

SmartStudy AI Design Document

1. System Workflows

- Assignment Planning Workflow

The Assignment Planning Workflow is designed to take input from the student and generate a structured, realistic study plan. The user provides details such as the course name, assignment description, due date, difficulty level, and available study time. This information is processed through a sequence of agents that break the assignment into smaller tasks, estimate workload, and create a day by day plan that fits within the student's schedule. The system also includes a calendar and Kanban-style dashboard to help students visually track their assignments. Each assignment is automatically placed on the calendar based on its due date and organized into color coded categories such as Overdue, Urgent, Soon, and On Track.

- Study Methods Recommendation Workflow

The Study Methods Recommendation Workflow provides personalized study strategies based on the student's learning preferences and current situation. The user answers a series of questions about their subject area, type of assignment, main challenges, learning style, available time, and energy levels. Using this input, the system generates tailored recommendations that include: The most effective study methods (ex: Pomodoro, active Recall, Practice Testing), explanations for why those methods fit the student, supportive coaching tips and guidance. This workflow focuses on helping students improve how they study, not just what they study. It provides insight into better habits, increases efficiency, and supports long term learning success

2. Agent Roles and Input/Output

—> **Assignment Planning Workflow Agents** - responsible for transforming student input into a structured and realistic study plan.

Planner Agent serves as the foundation of the workflow. It takes in user input such as course name, assignment details, due date, difficulty, and study availability. Its role is to analyze the assignment and break it into smaller tasks while estimating workload and available study days. The output is returned in a structured JSON format to ensure consistency.

- **Input:** Assignment details, due date, difficulty, study time
- **Output:** JSON including task breakdown, estimated hours, study days, and risk note

Scheduler Agent converts the structured output from the Planner into a readable study plan. It organizes tasks into a day by day schedule that fits within the student's available time and does not exceed the deadline. This agent focuses on clarity and usability.

- **Input:** Planner JSON output
- **Output:** Formatted study plan (summary, tasks, daily schedule, tips)

Critic Agent acts as a quality control layer. It reviews the draft plan and ensures that it is realistic, properly structured, and aligned with the student's constraints. If issues are found, the plan is revised and improved.

- **Input:** Planner output + Scheduler draft
- **Output:** Finalized and improved study plan

Reflection Agent provides explanation and guidance. It helps the student understand why the plan was structured in a certain way and highlights key risks or next steps.

- **Input:** Planner output + final plan
- **Output:** Explanation with insights and recommendations

—> **Study Methods Recommendation Workflow Agents** - This workflow focuses on improving how students study by generating personalized strategies based on their preferences and challenges.

The **Profile Agent** analyzes the student's responses, including subject area, learning style, challenges, and available time. It produces a structured summary of the student's needs and strengths. The **Method Agent** then uses this information to select the most effective study techniques and explain why they are appropriate. The **Planning Agent** converts these methods into a simple, actionable plan, and the **Coach Agent** provides supportive advice and encouragement.

- **Profile Agent Output:** student summary, main challenge, strengths
- **Method Agent Output:** recommended study methods and reasoning
- **Planning Agent Output:** step-by-step study plan
- **Coach Agent Output:** guidance, tips, and encouragement

3. How the Agents Communicate

Communication between agents is handled through a sequential pipeline controlled by the Flask backend. Each agent's output is passed directly to the next agent, ensuring a clear and organized flow of information. Structured outputs, particularly JSON, are used whenever consistency is required, while cleaned text outputs are used for user facing content. Utility functions are implemented to extract JSON from model responses and to clean and format text outputs, ensuring that data remains usable throughout the workflow.

- Structured data (JSON) is used for: Planner, Profile, Method, and Planning agents
- Cleaned text is used for: Scheduler, Critic, Reflection, and Coach agents

This combination ensures both accuracy/structured data and readability/formatted text. Utility functions are used to extract JSON and clean outputs before passing them along, ensuring consistency across the system.

4. Failure Handling and Low-Quality Output Control

API Failure Handling:

The system uses a retry mechanism when calling the Gemini API. It first attempts a primary model and, if necessary, falls back to a secondary model. Temporary errors such as service unavailability are retried automatically.

- API fallback: `generate_with_retry()` tries the primary Gemini model first, retries temporary failures, and then switches to the fallback model if needed.

Input Validation:

Before running any workflow, the system checks that:

- All required fields are completed
- Study hours and days are valid
- Dates are correctly formatted
- `validate_form_data()` checks missing fields, numeric ranges, and due-date parsing before generation begins.

Low-Quality Output Handling:

The Critic agent plays a key role in improving output quality by. The Critic agent acts as an internal reviewer and improves realism, deadline accuracy, and scanability before the plan is shown.

- Fixing unrealistic schedules
- Ensuring plans do not exceed deadlines
- Improving clarity and formatting
- Output cleanup: `clean_text()` removes repeated blank lines so plans and reflections are easier to read.

Data Integrity:

If structured outputs fail to parse correctly, the system stops the workflow to prevent invalid results. This ensures that only clean and usable data is processed and displayed.

- Format enforcement: Planner style agents are told to return only valid JSON in an exact format. If parsing fails, the app raises an error instead of saving bad data.

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SmartStudy AI

Student Productivity Dashboard

Dashboard

Assignment Board

Calendar

New Assignment

Study Methods

OVERVIEW

Overdue

0

Urgent

1

Soon

1

On Track

1

Total

3

NEW ASSIGNMENT

Create Assignment

Open a full-page form to enter details and generate a study plan.

Course —

Title —

Open Assignment Form

Student Productivity Dashboard

Multi-agent academic planning with organization, calendar tracking, and AI-generated study support.

Overdue: 0

Urgent: 1

Soon: 1

On Track: 1

ASSIGNMENTS

3

Total items currently tracked in your planner.

CALENDAR

March 2026

Every assignment added to the planner appears on the calendar automatically.

AI SYSTEM

Smart Planning Workflow

SmartStudy AI builds structured study plans, improves them, and adds helpful guidance for each assignment.

CALENDAR VIEW

March 2026

← Prev

Next →

SUN	MON	TUE	WED	THU	FRI	SAT
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21

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OVERVIEW

Overdue

0

Urgent

1

Soon

1

On Track

1

Total

3

NEW ASSIGNMENT

Create Assignment

Open a full-page form to enter details and generate a study plan.

Course —

Title —

Open Assignment Form

Assignment Board

Organized by Priority

Overdue

0

No assignments in this column yet.

Urgent

1

CIS 4730

Report

Urgent

DUE DATE

03/26/2026

DIFFICULTY

Easy

HOURS / DAY

5.0

DAYS / WEEK

1

1 day(s) left

View Study Plan

Why This Plan Works

Click to expand

EditDelete

Soon

1

CIS 4730

Essay

Soon

DUE DATE

03/29/2026

DIFFICULTY

Easy

HOURS / DAY

2.0

DAYS / WEEK

3

4 day(s) left

View Study Plan

Why This Plan Works

Click to expand

EditDelete

On Track

1

CIS 4400

Report

On Track

DUE DATE

04/08/2026

DIFFICULTY

Hard

HOURS / DAY

2.0

DAYS / WEEK

5

14 day(s) left

View Study Plan

Why This Plan Works

Click to expand

EditDelete

S

SmartStudy AI

Student Productivity Dashboard

Dashboard

Assignment Board

Calendar

New Assignment

Study Methods

OVERVIEW

Overdue0

Urgent1

Soon1

On Track1

Total3

PAGE FOCUS

Answer a few questions to get personalized study methods, a realistic action plan, and helpful guidance.

Study Methods

Personalized study strategies based on your subject, challenge, learning style, and available time.

PERSONALIZED INPUTS

Tell us about your study situation

Subject Area

Ex. Biology, Accounting, Programming, Marketing

What are you preparing for?

Select type

What is your biggest challenge?

Select all that apply

Select challenge(s)

How do you learn best?

Select all that apply

Select learning style(s)

How much time do you have today?

Select time

How do you prefer to study?

Select all that apply

Select preference(s)

When is your best energy time?

Select time

How confident do you feel right now?

Select level

Generate Study Methods

PERSONAL STUDY SUMMARY

Your Best Starting Direction

Fill out the form to get personalized study methods, tips, and a clear action plan.

STUDY PROFILE

How You Study Best

Your personalized profile will appear here.

RECOMMENDED METHODS

Best Methods for You

Recommended methods will appear here.

ACTION PLAN

Today's Smart Plan

Your action plan will appear here.

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SmartStudy AI
Student Productivity Dashboard

Dashboard

Assignment Board

Calendar

New Assignment

Study Methods

OVERVIEW

Overdue0

Urgent1

Soon1

On Track1

Total3

PAGE FOCUS

Answer a few questions to get personalized study methods, a realistic action plan, and helpful guidance.

Method Library

Focus

Pomodoro Technique

Study for 25 minutes, then take a 5-minute break. This helps you stay focused without burning out.

- Best for: procrastination and focus problems
- Use a timer and remove distractions first

Memory

Active Recall

Instead of rereading, close your notes and test what you actually remember.

- Best for: exams and memorization-heavy classes
- Use self-quizzes, flashcards, or blank-page recall

Retention

Spaced Repetition

Review information multiple times over several days instead of cramming everything at once.

- Best for: long-term memory
- Review earlier topics before they fade

Practice

Practice Testing

Use practice problems, sample questions, and quizzes to learn by doing.

- Best for: math, accounting, science, and exams
- Focus on weak areas, not just easy questions

Planning

Time Blocking

Assign study tasks to specific time slots so your work has a clear place in your day.

- Best for: busy schedules and time management
- Block your hardest task during your highest-energy hours

Review

Blurting Method

Write down everything you know from memory, then compare it to your notes to find gaps.

- Best for: checking true understanding
- Great before exams or presentations

QUICK TIPS

- Break large assignments into smaller tasks.
- Study hard subjects first when energy is highest.
- Do not wait until motivation appears — start small.
- Review class notes within 24 hours.

RECOMMENDED WORKFLOW

- Read the assignment carefully.
- Split it into 2–5 tasks.
- Schedule those tasks before the due date.
- Use active recall or practice testing to review.

BEST MATCH GUIDE

- Focus issues:** Pomodoro
- Memorization:** Active Recall + Spaced Repetition
- Problem-solving classes:** Practice Testing
- Busy schedule:** Time Blocking